

Bachelor's Thesis 202X



Evaluating Information Access Systems: An Undergraduate Thesis Demonstration

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Abstract

This document demonstrates the updated English undergraduate thesis template using an illustrative study of information access systems. It contains a research question, a proposed evaluation procedure, example tables, a mathematical definition, and a discussion of limitations. All numerical results are invented for typesetting purposes. The journal and conference references are also explicitly fictional examples, rather than evidence for the statements below.

The layout separates chapter-opening pages from continuation pages. Each chapter begins on a fresh page with a centred page number and no running header. Later pages retain alternating chapter and section headers. Chapters may begin on an odd or even page, so no empty page is inserted merely to reach a right-hand page. The first chapter includes two continuation pages to make both running header positions visible. The remaining chapters contain a continuation page for comparison. These demonstration page breaks can be removed when writing an actual thesis.

The example also demonstrates a table of contents, cross-references, acknowledgements, a bibliography and an appendix. Together, these elements provide a practical document for reviewing margins, heading hierarchy, caption placement and continuous page numbering.

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Chapter 1

Introduction

1.1 Background and Motivation

Information access systems help users locate material that is relevant to a question or task. A search interface may return a ranked list of documents, while another interface may combine retrieved passages into a short answer. In either case, evaluation begins with a clear account of the user need and the evidence that would satisfy it. A system that retrieves useful material for one type of question may perform differently when the vocabulary, topic, or expected level of detail changes.

This illustrative thesis considers how to compare two retrieval configurations under a shared evaluation procedure. The purpose of the example is to provide realistic paragraph lengths and document structure for checking the template. It does not report a completed experiment. Any statements about the example systems describe the proposed study design, and the numerical values shown later are invented. Keeping this distinction explicit also demonstrates how a thesis can separate an experimental plan from observed results.

1.2 Research Questions

The example asks whether an additional reranking stage improves the usefulness of the first few retrieved documents, and whether any improvement is consistent across different query groups. These are separate questions. An increase in an overall mean does not establish that every group benefits, because the mean can combine gains on some queries with losses on others. The evaluation must therefore retain query-level observations as well as summary values.

The main comparison uses the same collection, queries and relevance judgements for both configurations. Only the ranking procedure changes. This restriction makes the comparison easier to interpret, although it does not eliminate all sources of uncertainty. Later chapters discuss judgement coverage, parameter selection and the limits of a small evaluation collection.

1.3 Scope and Contributions

The intended contribution of this example study is an explicit and reproducible comparison procedure. A useful report should explain how the input collection was prepared, how queries were selected, which settings were held constant, and how per-query scores were aggregated. The reader should be able to trace a value in a results table back to the observations and choices that produced it. A single headline score is not a substitute for this account.

Three outputs are envisaged. The first is a documented baseline configuration. The second is a controlled variant that introduces a reranking stage. The third is an analysis that considers both average performance and variation across queries. These outputs are described as a plan, not as completed research contributions. In a real thesis, this section should be rewritten to reflect what was actually implemented and established by the evidence.

1.3.1 Assumptions

The example assumes that each query has a fixed information need and that the available judgments are used consistently for both systems. It also assumes that the collection remains unchanged during the experiment. These assumptions simplify the comparison, but they limit the conclusions: an offline experiment does not directly establish how people would use the system in a live setting. The distinction should remain visible when reporting the results.

1.3.2 Reading the Example

The document includes both numbered chapters and unnumbered back matter. Tables and figures have independent numbering, and references to them are resolved automatically. For example, Chapter 4 describes the experimental design, while Appendix A lists the information needed to reproduce it. These references should remain correct after material is added or removed.

1.4 Organisation of the Thesis

Chapter 2 explains how to organise related work around the decisions made in an evaluation. Chapter 3 then defines the illustrative comparison and its aggregation rule. The experimental chapter provides example tables, and Chapter 5 shows how to discuss an invented result without presenting it as an empirical finding. The final chapters distinguish conclusions from possible extensions.

This organisation places the study design before the results. That order helps the reader determine which decisions were part of the original procedure and which interpretations arose afterwards. A complete thesis may need additional chapters for implementation, datasets or user studies. The template does not require every project to follow the particular chapter structure used here.

1.5 Layout Inspection

On this continuation page, the running header should identify the current chapter or section, according to page parity. The footer should display the page number in the same horizontal position as on the chapter-opening page. The opening page itself should have no running header and no header rule. This first chapter has three pages so that both ordinary header positions can be inspected without changing the source.

The next chapter begins on the immediately following page. Its page number may be even or odd; no empty page is required between chapters. The forced page breaks within these sample chapters are solely for demonstrating page styles. During normal writing, paragraphs flow automatically and each `\chapter` command starts a new page on its own.

Chapter 2

Related Work

2.1 Organising the Literature

A related-work chapter should help the reader understand why the study makes particular methodological choices. One possible organisation separates retrieval methods, evaluation resources and analysis procedures. Another follows the research questions directly. The choice should make comparisons clear rather than simply reproduce the order in which papers were read. For each relevant study, the discussion should identify its problem, evidence, and relationship to the present work.

The citations in this demonstration illustrate formatting only. The laboratory website entry [1] demonstrates a web address in a bibliography. The journal entry [2] and conference entry [3] are fictional and are not sources for the methodological discussion. In a real thesis, replace these entries with verified publications and ensure that each citation supports the sentence in which it appears.

2.2 Comparing Experimental Settings

Two studies may use similar ranking methods while evaluating substantially different tasks. Differences in collection size, query selection, judgement coverage or metric definition can affect the interpretation of their scores. A careful comparison therefore reports the experimental setting alongside the method. It should avoid treating values from unrelated collections as if they were measurements on the same scale under identical conditions.

A compact comparison table can summarise these settings, but the surrounding text still needs to explain their significance. For instance, a collection containing short factual queries may not reveal the same behaviour as a collection of exploratory requests. Such differences can motivate an additional experiment without implying that either earlier study was flawed.

2.3 Identifying a Research Gap

A research gap should be stated in terms that can be investigated. Saying that a topic has received limited attention is less specific than identifying an unresolved comparison or an untested assumption. In this example, the proposed gap concerns whether a change in average retrieval quality is accompanied by consistent behaviour across query groups. That statement leads to observable quantities and a concrete analysis plan.

The scope of the claim should match the literature that was examined. A thesis can explain its search strategy, inclusion criteria and stopping point without claiming that every relevant publication has been found. It can also separate a practical implementation gap from a conceptual limitation. These distinctions help prevent an overly broad novelty claim from obscuring a useful, narrower contribution.

2.4 Position of the Present Example

The proposed comparison uses an established baseline as a reference point and changes a single component. The role of related work is to justify that baseline, explain reasonable alternatives and identify possible confounding factors. A stronger or newer method is not automatically the most informative baseline if its data preparation or computational requirements cannot be matched.

The next chapter translates these considerations into a procedure. It defines the inputs, the two configurations and the aggregation of per-query scores. This transition connects the literature review with the actual experiment, so that the reader can see how earlier observations influenced the study design rather than encountering an unrelated list of publications.

Chapter 3

Proposed Methods

3.1 System Overview

The example compares a baseline retrieval configuration with a variant that reranks an initial candidate set. Both configurations receive the same query and search the same collection. The baseline returns its initial ordering, whereas the variant applies an additional scoring function to the candidates. Documents outside that candidate set cannot be promoted by the reranker, which makes candidate selection an important part of the experimental description.

The implementation should record the candidate-set size, preprocessing rules, and treatment of tied scores. These details can change the output even when the high-level method name remains the same. Configuration files and software versions should be saved with the result files. The appendix provides an example checklist for this record, rather than prescribing a particular software framework.

3.2 Aggregation of Query Scores

Let s_i denote a selected evaluation score for query i , with n queries included in the comparison. The mean score is

$$\bar{s} = \frac{1}{n} \sum_{i=1}^n s_i. \quad (3.1)$$

Equation (3.1) assigns equal weight to each query. A different weighting scheme would answer a different question and should be stated explicitly. The metric used to obtain s_i must also be defined; the aggregation formula alone does not specify how relevance, rank or unjudged documents are handled.

The example retains both the mean and the individual scores. This allows the analysis to distinguish a small improvement shared by many queries from a large improvement concentrated in only a few cases. Neither pattern can be reconstructed from the mean alone.

3.3 Paired Comparison

For each query, define the difference between the variant and baseline scores using the same relevance judgements. Pairing the observations removes ambiguity about which queries contributed to each system's mean. If a query is excluded, the reason should be documented and the exclusion should be applied consistently to both configurations.

A useful result file contains the query identifier, both scores and their difference. Additional fields may describe the query group or judgement coverage. These annotations support later analysis, but they should not be chosen merely because they produce a favourable comparison. Decisions about groups and exclusions belong in the experimental design wherever possible.

3.3.1 Quality Checks

Before aggregation, check for duplicate identifiers, missing outputs and unexpected score ranges. The comparison should fail visibly when a required system output is absent. Replacing an absent output with a score of zero may be appropriate in some protocols, but it must be an explicit evaluation rule, not an accidental consequence of a file-reading routine.

3.4 Reproducibility

The final report should distinguish deterministic settings from operations that depend on a random seed. If repeated runs are used, explain what changes between runs and how the resulting observations are summarised. Saving a seed without recording the software and data versions is not a complete description of the experimental environment.

The procedure should be checked on a small example before applying it to the full collection. Manually inspecting a few query-level calculations can reveal mismatched identifiers or inconsistent cutoffs. This check complements, rather than replaces, automated validation of the complete result files.

Chapter 4

Experiments

4.1 Experimental Configuration

This chapter illustrates how to describe an experimental setup. The values in Table 4.1 are hypothetical configuration values, not a record of an experiment performed for this template. A completed thesis should replace them with the actual settings and explain which choices were fixed before examining the evaluation results.

Table 4.1 Illustrative configuration; no experiment was executed

Component	Example setting
Candidate set	100 documents per query
Evaluation cutoff	First 10 results
Configurations	Baseline and reranked variant
Aggregation	Equal weight per query
Output record	One paired score per query

4.2 Collection and Queries

The collection description should identify the document source, any filtering steps and the version used in the evaluation. Query selection deserves the same level of care. Queries used during development should be distinguished from those reserved for the final comparison, and any overlap should be reported rather than left implicit.

Table 4.2 uses invented group sizes to demonstrate a second table and its cross-reference. A real report would explain how each group was defined and why the distinction is relevant to the research question. It would also state whether group membership was assigned before the system outputs were examined.

Table 4.2 Invented query groups for the formatting example

Group	Development queries	Evaluation queries
Short requests	20	40
Detailed requests	20	40
All requests	40	80

4.3 Execution Procedure

Run both configurations with the same prepared inputs and retain their complete ranked outputs. The evaluation stage should read those outputs without altering the rankings. Separating retrieval from scoring makes it possible to inspect a result file, correct an evaluation bug and recompute scores without silently changing the retrieval configuration at the same time.

Each run should have an identifier that connects its configuration, output and log. A run log can record completion status and unexpected conditions, while the configuration file records intended settings. If a run fails or produces an incomplete output, resolve the issue before presenting a comparison as a complete experiment. The final report should describe consequential deviations from the original procedure.

4.4 Analysis Plan

The first analysis compares mean query scores. The second examines the distribution of paired differences, and the third summarises the predefined query groups. These analyses answer related questions, but none should be used to imply more than it measures. A group summary based on a small number of queries may be informative while remaining highly uncertain.

Qualitative inspection can complement these summaries. Select examples using a stated rule, such as the largest positive and negative changes, and inspect the retrieved documents for plausible explanations. Such examples illustrate behaviour; they do not estimate how often that behaviour occurs in the broader query population. The next chapter uses a small invented result table to show how these distinctions can be written clearly.

Chapter 5

Results and Discussions

5.1 Illustrative Results

All values in Table 5.1 are invented for layout inspection. They must not be interpreted as measured performance. In this numerical example, the variant has a larger mean in both groups, but the difference is larger for short requests. Because the groups have equal size in the example design, the overall means are the simple averages of the two group means.

Table 5.1 Invented scores used only to demonstrate results formatting

Query group	Baseline	Variant	Difference
Short requests	0.420	0.460	+0.040
Detailed requests	0.380	0.390	+0.010
Overall	0.400	0.425	+0.025

A real discussion would report the uncertainty of the comparison and connect it to the research questions. It should avoid describing a numerical increase as a reliable improvement without the evidence needed for that statement. The size of a change also needs practical interpretation: a difference may be observable in an offline score without noticeably changing the usefulness of the system for its intended users.

5.2 Figure Example

Figure 5.1 demonstrates a centred image with its caption below it. The image is the existing template logo and conveys no experimental result. In a completed thesis, replace it with a relevant figure and explain what the reader should learn from it in the surrounding text.

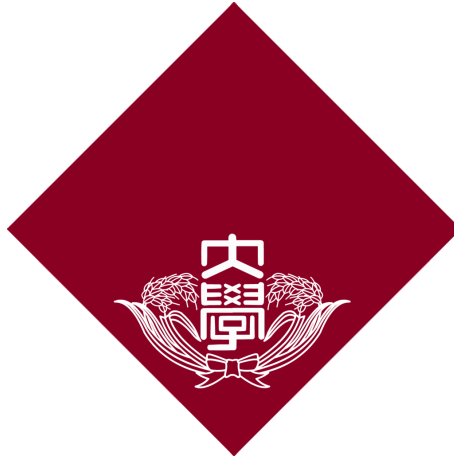


Fig. 5.1 Waseda University logo, included as a formatting example

5.3 Interpreting Differences

An aggregate result can conceal different query-level patterns. Some queries may benefit from a reranking stage while others lose a relevant document near the top of the list. Inspecting the paired scores helps identify these cases, but the interpretation should be grounded in the retrieved material rather than inferred from the sign of the score difference alone.

The discussion should distinguish observations from possible explanations. For example, a query might contain terminology that appears in several unrelated contexts. A retrieved document can reveal such an ambiguity, but that single case does not establish a general failure mode. Additional cases or a targeted experiment would be needed to support a broader claim.

5.4 Limitations

The illustrative design has several limitations. A fixed collection may not reflect changes in language or document availability. Incomplete relevance judgements may leave some useful documents unrecognised by the evaluation. A small query set may also give an unstable picture of group-level behaviour. Each limitation affects the scope of the conclusion in a different way.

A useful limitations section explains those effects rather than merely listing potential problems. It should identify which conclusions remain supported, which are tentative and which would require a new experiment. Since this chapter contains no real observations, its only conclusion is about how a results discussion can be structured. The following chapter preserves that boundary when summarising the example.

Chapter 6

Conclusion

6.1 Summary of the Example

This demonstration has described an illustrative comparison between a baseline retrieval configuration and a reranked variant. The proposed procedure holds the collection and queries fixed, retains paired query scores and examines both overall and group summaries. It is a worked example of thesis structure, not evidence that a particular retrieval method outperforms another.

The document connects research questions to an experimental plan and then to a discussion of possible results. That connection is useful in a completed thesis because it lets the reader assess whether the reported evidence answers the questions posed at the beginning. A conclusion should revisit those questions directly instead of introducing an unrelated set of claims.

6.2 Reporting Contributions

A final thesis should distinguish a software artifact, a methodological contribution and an empirical finding. Implementing a system establishes that an artifact exists; evaluating it under a documented protocol provides a different kind of evidence. Neither automatically demonstrates that the approach generalises beyond the setting that was studied.

For this template, the concrete artifact is the formatted document itself. It demonstrates chapter openings with visible page numbers, continuation pages with running headers, and uninterrupted chapter transitions on either page parity. Tables, a figure, citations and an appendix provide additional examples for reviewing the surrounding document structure.

6.3 Boundaries of the Conclusions

A conclusion should state the conditions under which its claims hold. Those conditions may include the collection version, query population, relevance judgements and computational constraints. Reporting them does not weaken an otherwise sound result; it helps the reader understand what would need to remain comparable when applying the finding elsewhere.

The invented scores in this document support no empirical claim. Their role is to exercise table alignment, decimal formatting and cross-references. Similarly, the fictional references demonstrate bibliography syntax without establishing a literature basis for the study. Both should be replaced before the template is used to report actual research.

6.4 Closing Remarks

The final paragraph of a thesis can explain why the established contribution matters while keeping its scope precise. It should follow from the evidence already presented rather than rely on a promise that future work will resolve all remaining issues. Specific possible extensions belong in the next chapter, where their motivation and requirements can be explained separately.

For layout inspection, compare this page with the preceding chapter-opening page. Both should display the same centred footer style. Only this continuation page should have a running header and a thin horizontal rule above the text. The next chapter starts on the next page, without an intervening empty leaf.

Chapter 7

Future Work

7.1 Broader Evaluation

One possible extension is to evaluate additional query collections with clearly different characteristics. Such an extension would test how sensitive the comparison is to the initial study setting. It should preserve enough of the original protocol to make the new evidence interpretable while explaining any changes required by the additional data.

A broader collection is not automatically a better experiment. Its relevance judgements, sampling procedure and documentation still determine what can be learned from it. Before expanding the study, identify the uncertainty that the new collection is intended to reduce. This makes the extension a targeted investigation rather than simply a larger repetition of the same procedure.

7.2 User-Centred Analysis

Another extension would examine how people interact with the retrieved results. An offline score represents selected aspects of retrieval quality, whereas a user study can investigate task completion, interpretation or effort under a specified interface. The study design would need to define the tasks and outcomes carefully so that the observations answer a focused question.

The interface, instructions and participant population would become part of the experimental setting. These elements should be documented alongside the retrieval configuration. A user study should not be presented as a direct replacement for an offline comparison; the two approaches can provide evidence about different aspects of the same system.

7.3 Efficiency and Deployment

A further direction is to examine the computational cost of the reranking stage. Relevant measurements might include response time, memory use and the number of candidates processed. The evaluation should state how these values were measured and which parts of the pipeline were included. A timing result without the corresponding execution conditions is difficult to interpret.

The eventual deployment setting may impose a different balance between ranking quality and computational cost. A small improvement may be useful in one application but insufficient to justify added delay in another. Establishing that balance requires a concrete account of the task and its constraints, rather than a universal claim about the best configuration.

7.4 Prioritising the Next Step

A practical research plan should identify which extension would be most informative and what resources it requires. The next step could be a targeted error analysis, an additional collection or a measurement of runtime cost. Its value depends on which uncertainty currently limits the interpretation of the study, not simply on how many new components it introduces.

This completes the numbered chapters of the example. Acknowledgements and references follow as unnumbered chapters, each with a visible page number on its opening page. The appendix then demonstrates lettered chapter numbering. These transitions are included so that the same page-style rules can be checked across the complete thesis structure.

Acknowledgements

This page is a formatting example for acknowledgements. In a completed thesis, it should identify the people and organisations whose contributions supported the work. The wording should distinguish academic advice, technical assistance, financial support and personal encouragement where appropriate.

Replace this paragraph with your own acknowledgements. Do not retain invented funding information or imply that an individual endorsed the conclusions of your thesis. This unnumbered chapter is included in the table of contents and uses the same opening-page style as the numbered chapters: a visible footer page number, with no running header or horizontal header rule.

References

- [1] RSL. The sakai laboratory. <http://www.f.waseda.jp/tetsuya/>. Replace with the current URL and your actual access date.
- [2] Jane Doe and John Roe. Example journal article: Replace with a real reference. *Example Journal*, 1(1):1–10, 2024.
- [3] Jane Doe and Alex Smith. Example conference paper: Replace with a real reference. In *Proceedings of the Example Conference*, pages 11–20, 2024.

Appendix A

Supplementary Details

A.1 Reproduction Record

An appendix can preserve details that are necessary for reproduction but would interrupt the main discussion. For the illustrative study, this includes the collection version, query identifiers, judgement files and the settings used for each configuration. The record should connect these inputs to the result files described in Chapter 4.

A concise checklist helps the reader locate the required material:

1. Record the exact collection and query versions.
2. Save preprocessing and retrieval settings for both configurations.
3. Retain ranked outputs and paired query-level evaluation scores.
4. Document exclusions and unexpected conditions.
5. State the command or procedure used to reproduce each reported table.

An appendix should remain readable on its own while making its relationship to the main text clear. Cross-references are preferable to manually typed page numbers because the document may change during revision. This appendix is labelled with a letter, and its sections use the corresponding letter prefix.

A.2 Checking the Compiled Document

This continuation page provides one final example of the running header and centred footer. The opening page of Appendix A should show its lettered title and a page number, but no running header. Its entry in the table of contents should point to that opening page rather than to this continuation.

When replacing the sample content, inspect the compiled document after major structural changes. Long headings, large tables and figures close to a page boundary can change pagination. Those changes are normal, but captions and cross-references should remain associated with the correct content. Remove the demonstration-only page breaks to allow the actual thesis to flow naturally.

The cover and abstract retain their existing unnumbered appearance. The table of contents and all pages after it display page numbers. Page counting continues from the start of the document, so the first visible number need not be one. This example does not introduce a separate Roman-numeral sequence for front matter or restart Arabic numbering at the first chapter.